Northwind Shipment Analysis Data Warehouse Project Documentation

---

1. Project Overview

Introduction

The Northwind Shipment Analysis project involves building a data warehouse to enable advanced business intelligence focused on shipment and delivery performance. The goal is to transform the normalized Northwind transactional database into a dimensional star schema that supports efficient analytics in tools such as Power BI.

The Northwind database is not designed for analytical workloads because it stores data in normalized operational tables such as Orders and Shippers. This project redesigns the structure into a star schema to support performance analysis and reporting.

Objectives

- Design and implement a star schema for shipment and delivery performance
- Load Northwind operational data into a structured data warehouse
- Create Power BI dashboards to analyze shipping KPIs
- Enable insights into shipping delays, carrier performance, and delivery efficiency

Business Value

This project provides value by enabling organizations to:

- Analyze shipping efficiency over time
- Compare performance across different shipping carriers
- Identify delayed or late shipments
- Improve operational logistics and customer satisfaction

---

2. Team Structure

Team Members and Roles

(Insert roles if required)

- Data Warehouse Designer: Responsible for dimensional modeling and schema design
- ETL Developer: Responsible for data extraction, transformation, and loading into Snowflake
- BI Developer: Responsible for building Power BI dashboards and visualizations

Project Charter

The mission of this project is to build a scalable and efficient data warehouse that transforms Northwind operational data into meaningful shipment performance insights for decision-making.

---

3. Project Plan

Timeline and Milestones

- Milestone 1: Bus Matrix and High-Level Design
- Milestone 2: Dimensional Modeling and ETL Implementation
- Milestone 3: Power BI Dashboard Development and Presentation

Tasks and Deliverables

- Create bus matrix defining business processes and dimensions
- Design star schema (fact and dimension tables)
- Implement ETL pipeline in Snowflake
- Build analytical queries for validation
- Develop Power BI dashboard with KPIs and insights

---

4. Data Overview

Source Data Description
The Northwind dataset includes operational transactional data used for shipping analysis. The main tables used include:

Orders Table:
Contains order-level data including OrderID, OrderDate, RequiredDate, ShippedDate, ShipVia, ShipCountry, ShipRegion, ShipCity, and Freight.

Shippers Table:
Contains shipping carrier information such as ShipperID, CompanyName, and Phone.

Data Preparation
The raw data was extracted and loaded into Snowflake staging tables and then transformed into structured dimensional tables.

Data Challenges

- Handling missing ShippedDate values for incomplete shipments
- Ensuring correct date transformations for analysis
- Establishing correct relationships between Orders and Shippers

---

5. Dimensional Modeling

Bus Matrix

Business Process: Shipment and Delivery Performance

Business Process	Date	Shipper	Customer Location	Order
Shipment Analysis	Yes	Yes	Yes	Yes

Star Schema Design

Fact Table: FACT_SHIPMENT
The fact table represents one row per order shipment and contains measurable shipping performance attributes.

Dimensions:

- DIM_DATE
- DIM_SHIPPER
- DIM_CUSTOMER (optional geographic attributes)

---

6. Data Warehouse Implementation

Technical Architecture
The data warehouse was implemented using Snowflake with a layered architecture consisting of:

- Raw Layer: Stores extracted Northwind data
- Staging Layer: Temporary transformation layer
- Star Schema Layer: Fact and dimension tables for analytics

ETL Pipeline Design

Extract:
Data was extracted from Northwind Orders and Shippers tables.

Transform:

- Calculated shipping duration (ShippedDate - OrderDate)
- Created surrogate keys for date dimension
- Derived late shipment flags

Load:
Data was loaded into dimension and fact tables in Snowflake.

Snowflake Implementation Queries

Database Setup:

```
CREATE DATABASE IF NOT EXISTS northwind_db;
CREATE SCHEMA IF NOT EXISTS raw_orders;
USE DATABASE northwind_db;
USE SCHEMA raw_orders;
```

Stage Creation:

```
CREATE OR REPLACE STAGE northwind_stage
FILE_FORMAT = (
TYPE = 'CSV'
SKIP_HEADER = 1
);
```

Raw Orders Table:

```
CREATE OR REPLACE TABLE raw_orders (
OrderID STRING,
OrderDate DATE,
RequiredDate DATE,
ShippedDate DATE,
ShipVia INT,
ShipCountry STRING,
ShipRegion STRING,
ShipCity STRING,
Freight FLOAT
);
```

Data Load:

```
COPY INTO raw_orders
FROM @northwind_stage/orders.csv
FILE_FORMAT = (
TYPE = 'CSV'
SKIP_HEADER = 1
)
ON_ERROR = 'CONTINUE';
```

Dimension Tables

DIM_DATE:
Created using distinct OrderDate values and includes year, quarter, month, and day breakdowns.

DIM_SHIPPER:
Derived from Shippers table and includes shipper identifiers and company information.

DIM_CUSTOMER:
Derived from order-level geographic attributes such as ShipCountry and ShipRegion.

Fact Table: FACT_SHIPMENT
The fact table integrates order and shipping data and includes calculated measures.

Key fields include:

- order_id
- order_date_id
- ship_date_id
- required_date_id
- shipper_id
- ship_country
- ship_region
- freight
- shipping_duration
- late_flag

---

7. Business Intelligence (BI) Implementation

Dashboard Overview
The Power BI dashboard was designed to provide insights into shipment performance, including:

- Average shipping duration trends over time
- Carrier performance comparison
- Late delivery analysis
- Shipping performance by country and region

Key Metrics and Insights

- Shipping Duration: Measures delivery efficiency
- Late Delivery Rate: Percentage of delayed shipments
- Carrier Comparison: Performance differences between shippers
- Geographic Analysis: Regional shipping performance trends

Sample Analytical Queries

Shipping Trend Over Time:
```
SELECT d.year, d.month, AVG(f.shipping_duration)
FROM fact_shipment f
JOIN dim_date d ON f.order_date_id = d.date_id
GROUP BY d.year, d.month;
```

Shipper Performance:
```
SELECT shipper_id, AVG(shipping_duration)
FROM fact_shipment
GROUP BY shipper_id;
```

Late Shipments:
```
SELECT SUM(late_flag), COUNT(*)
FROM fact_shipment;
```

---

8. Challenges and Solutions

Challenges:

- Missing or null shipping dates
- Establishing correct date relationships
- Ensuring accurate aggregation in Power BI

Solutions:

- Applied conditional logic for late shipment detection
- Created surrogate keys for date consistency
- Established star schema relationships for proper filtering

---

9. Reflection and Next Steps

This project demonstrated how operational data can be transformed into a structured analytical model using dimensional design principles. It improved understanding of ETL pipelines, star schema architecture, and BI dashboard development.

Future improvements include:

- Adding real-time data ingestion
- Expanding to multi-fact star schemas
- Enhancing predictive analytics for shipment delays

TABLE DEFINITIONS AND QUERIES

NORTHWIND_DB / RAW_ORDERS

Schema · SYSADMIN · 2 days ago

Schema Details | Tables | Stages

6 Tables

Search | All Tables

NAME	T...	CLAS...	OWNER	R...	B...	CRE...
DIM_DATE	Ta...	—	SYSADMIN	494	4.0...	2 da...
DIM_LOCATION	Ta...	—	SYSADMIN	55	2.5...	2 da...
DIM_SHIPPER	Ta...	—	SYSADMIN	3	1.5...	2 da...
FACT_SHIPMENT	Ta...	—	SYSADMIN	636	18....	2 da...
ORDERS	Ta...	—	SYSADMIN	636	27....	2 da...
SHIPPERS	Ta...	—	SYSADMIN	3	1.5...	2 da...

```sql
create or replace TABLE NORTHWIND_DB.RAW_ORDERS.DIM_DATE (
    DATE_ID NUMBER(38,0),
    FULL_DATE DATE,
    YEAR NUMBER(4,0),
    MONTH NUMBER(2,0),
    DAY NUMBER(2,0)
);
```

Show less

```sql
create or replace TABLE NORTHWIND_DB.RAW_ORDERS.DIM_LOCATION (
    SHIPCOUNTRY VARCHAR(16777216),
    SHIPREGION VARCHAR(16777216),
    SHIPCITY VARCHAR(16777216)
);
```

```sql
create or replace TABLE NORTHWIND_DB.RAW_ORDERS.DIM_SHIPPER (
    SHIPPERID NUMBER(38,0),
    COMPANYNAME VARCHAR(16777216),
    PHONE VARCHAR(16777216)
);
```

```sql
create or replace TABLE NORTHWIND_DB.RAW_ORDERS.FACT_SHIPMENT (
    ORDER_ID NUMBER(38,0),
    ORDER_DATE_ID NUMBER(38,0),
    REQUIRED_DATE_ID NUMBER(38,0),
    SHIP_DATE_ID NUMBER(38,0),
    SHIPPER_ID NUMBER(38,0),
    SHIPCOUNTRY VARCHAR(16777216),
    SHIPREGION VARCHAR(16777216),
    SHIPCITY VARCHAR(16777216),
    FREIGHT FLOAT,
    SHIPPING_DURATION NUMBER(9,0),
    LATE_FLAG NUMBER(1,0)
);
```

Show less